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MATLAB Drive

Files

Name

discretetimesignal.mlx

FirstStepsInMATLAB.mlx

sinewave.mlx

Workspace

Name	Value	Size	Class
n	1×21 double	1×21	double
x	1×21 double	1×21	double

sinewave.mlx × discretetimesignal.mlx ×

/MATLAB Drive/discretetimesignal.mlx

```

2      clc;
3      clear;
4      close all;
5      % Discrete-time index
6      n = -10:10;
7      % Discrete-time sinusoidal signal
8      x = sin(0.2*pi*n);
9      % Plotting
10     figure;
11     stem(n, x, 'filled');
12     grid on;
13     title('Discrete Time Sinusoidal S:
14     xlabel('Sample Index (n)');
15

```

Discrete Time Sinusoidal Signa

Copilot Chat

Copilot

Hello! Copilot is here to answer your questions, help write and explain code, and even identify code issues. Learn more about [Copilot](#).

Copilot sometimes produces output that seems accurate but is not. Validate generated output before use. Share feedback on the output to help improve the responses.

discretet...gnal.mlx

Ask Copilot

Validate generated output before use.

Command Window

New to MATLAB? See resources for [Getting Started](#).

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FILE NAVIGATE TEXT CODE ANALYZE SECTION RUN

C:\Users\ADMIN\Documents\MATLAB

Files

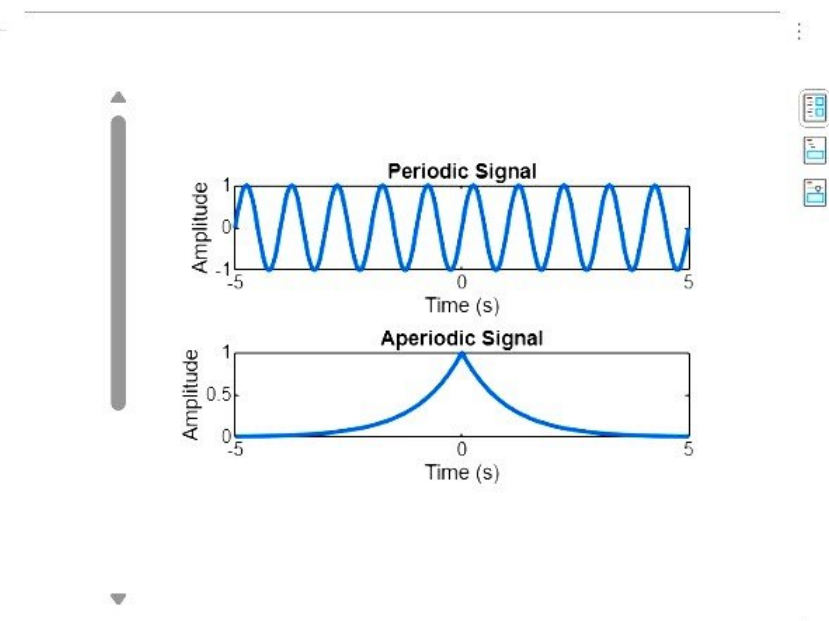
- Discrete.mlx
- Periodic.mlx
- sinewave.mlx
- triangleandsawtooth.mlx
- Trianglewave.mlx
- untitled.m

Workspace

Name	Value	Size	Class
t	1×1001 dou...	1×1001	double
x1	1×1001 dou...	1×1001	double
x2	1×1001 dou...	1×1001	double

```

1 % Periodic and Aperiodic Signals
2 clc;
3 clear;
4 close all;
5 % Time axis
6 t = -5:0.01:5;
7 % Periodic Signal (Sine Wave)
8 x1 = sin(2*pi*t);
9 % Aperiodic Signal (Exponential Decay)
10 x2 = exp(-abs(t));
11 % Plotting
12 figure;
13 subplot(2,1,1);
14 plot(t,x1,'LineWidth',2);
15 grid on;
16 title('Periodic Signal');
  
```



Command Window

New to MATLAB? See resources for [Getting Started](#).

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FILE NAVIGATE TEXT CODE ANALYZE SECTION RUN

C:\Users\ADMIN\Documents\MATLAB

Files

Name

- Discrete.mlx
- evenodd.mlx
- Periodic.mlx
- sinewave.mlx
- triangleandsawtooth.mlx
- Trianglewave.mlx
- untitled.m

Workspace

Name	Value	Size	Class
t	1×101 double	1×101	double
x	1×101 double	1×101	double
x_recons...	1×101 double	1×101	double
x_rev	1×101 double	1×101	double
xe	1×101 double	1×101	double
xo	1×101 double	1×101	double

sinewave.mlx × Discrete.mlx × Periodic.mlx × evenodd.mlx ×

C:\Users\ADMIN\Documents\MATLAB\evenodd.mlx

```

1      % Verification of Even and Odd Signal Decomposition
2      clc;
3      clear;
4      close all;
5      % Time axis
6      t = -5:0.1:5;
7      % Original Signal
8      x = t.^3 + t;
9      % Time-reversed Signal
10     x_rev = (-t).^3 + (-t);
11     % Even Component
12     xe = (x + x_rev)/2;
13     % Odd Component
14     xo = (x - x_rev)/2;
15     % Reconstruction of Original Signal

```

Command Window

New to MATLAB? See resources for [Getting Started](#).

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